

IMAT QUESTION BANK: Maths, Physics and Logic (Medium-Hard)

95 questions · 158 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · Question bank

Section A - Algebra, functions and logarithms (Q1-16)

- Given that $\log_{10} 2 \approx 0.30$ and $\log_{10} 3 \approx 0.48$, what is the value of $\log_{10} 7.5$?
A) 0.18 **B)** 0.58 **C)** 0.88 **D)** 1.18 **E)** 1.88
- Solve the equation $\log_3(x + 1) + \log_3(x - 1) = 1$.
A) $x = \pm 2$ **B)** $x = 2$ only **C)** $x = -2$ only **D)** $x = 4$ **E)** $x = 1$
- What is the value of $\log_8 32$?
A) 2 **B)** $3/2$ **C)** 3 **D)** $4/3$ **E)** $5/3$
- What is the solution of $\log_2 x + \log_4 x = 6$?
A) 4 **B)** 8 **C)** 16 **D)** 32 **E)** 64
- Which interval is the solution set of $(x - 1)(x + 2)(x - 4) > 0$?
A) $-2 < x < 4$
B) $1 < x < 4$
C) $x < -2$ or $x > 4$
D) $-2 < x < 1$ or $x > 4$
E) $x < -2$ or $1 < x < 4$
- What is the solution set of the inequality $(x - 3)/(x + 2) \leq 0$?
A) $-2 < x \leq 3$
B) $x \leq 3$
C) $x < -2$ or $x \geq 3$
D) $-2 \leq x \leq 3$
E) $-2 < x < 3$
- What is the solution set of $|2x - 5| < 7$?
A) $1 < x < 6$ **B)** $-6 < x < -1$ **C)** $-6 < x < 6$ **D)** $-1 < x < 6$ **E)** $x < 6$
- What is the solution set of the inequality $x^2 < x$?
A) $0 < x < 1$ **B)** $x < 0$ **C)** $x < 0$ or $x > 1$ **D)** $x < 1$ **E)** $x > 1$
- Which expression is equal to $\sqrt{7 + 4\sqrt{3}}$?
A) $4 + \sqrt{3}$ **B)** $2 + \sqrt{3}$ **C)** $\sqrt{7} + \sqrt{3}$ **D)** $2\sqrt{3} + 1$ **E)** $1 + \sqrt{6}$
- What is the value of $27^{(2/3)} + 16^{(3/4)}$?
A) 13 **B)** 17 **C)** 21 **D)** 26 **E)** 30
- If $2^a = 3$ and $2^b = 5$, which of the following is equal to 60?
A) $2^{(a + b)}$ **B)** $2^{(2a + b)}$ **C)** $2^{(a + b + 2)}$ **D)** $2^{(a + 2b)}$ **E)** $2^{(ab)}$
- For a school show, adult tickets cost 12 euro and child tickets cost 7 euro. In total 180 tickets were sold, and the takings were 1730 euro. How many adult tickets were sold?
A) 84 **B)** 86 **C)** 90 **D)** 94 **E)** 96
- The sum of two numbers is 45 and their difference is 13. What is the larger number?
A) 13 **B)** 16 **C)** 29 **D)** 32 **E)** 45
- A laboratory technician mixes a 20% saline solution with a 50% saline solution to obtain 6 dm³ of a 30% saline solution. What volume of the 50% solution is needed?
A) 1 dm³ **B)** 2 dm³ **C)** 2.5 dm³ **D)** 3 dm³ **E)** 4 dm³

15. The function f is defined by $f(x) = (x + 1)/(x - 1)$ for $x \neq 1$. What is $f(f(3))$?

- A) $1/2$ B) 2 C) 3 D) 4 E) 9

16. The function f is defined by $f(x) = (2x + 1)/(x - 3)$ for $x \neq 3$. Which of the following is the inverse function $f^{-1}(x)$?

- A) $(3x - 1)/(x + 2)$
B) $(x + 3)/(2x - 1)$
C) $(2x + 1)/(x - 3)$
D) $(3x + 1)/(x + 2)$
E) $(3x + 1)/(x - 2)$

Section B - Geometry, trigonometry and graphs (Q17-28)

17. A solid wooden block has the shape of a right circular cylinder of radius 6 cm and height 9 cm. A conical hole with the same base radius and the same height is drilled out of the block. What fraction of the cylinder's volume remains?

- A) $1/3$ B) $1/2$ C) $2/3$ D) $3/4$ E) $8/9$

18. A circle of radius 5 cm is inscribed in a square. Taking $\pi = 3.14$, what is the area of the part of the square that lies outside the circle?

- A) 21.5 cm^2 B) 39.25 cm^2 C) 78.5 cm^2 D) 100 cm^2 E) 121.5 cm^2

19. An equilateral triangle has sides of length 12 cm. What is its area?

- A) $36\sqrt{3} \text{ cm}^2$ B) 72 cm^2 C) $72\sqrt{3} \text{ cm}^2$ D) 144 cm^2 E) $144\sqrt{3} \text{ cm}^2$

20. A closed cylindrical tin has radius 3 cm and height 8 cm. What is its total surface area, ends included?

- A) $42\pi \text{ cm}^2$ B) $48\pi \text{ cm}^2$ C) $57\pi \text{ cm}^2$ D) $66\pi \text{ cm}^2$ E) $72\pi \text{ cm}^2$

21. What is the angle between the hour hand and the minute hand of a clock at 3:40? (The hour hand turns at 0.5° per minute, the minute hand at 6° per minute.)

- A) 100° B) 110° C) 130° D) 150° E) 230°

22. Between 7:00 and 8:00 the hands of a clock are exactly 100 degrees apart at 7:20. At what later time in that hour are they exactly 100 degrees apart again? (The hour hand turns at 0.5 degrees per minute, the minute hand at 6 degrees per minute.)

- A) 7:52 and $8/11$ minutes
B) 7:54 and $6/11$ minutes
C) 7:56 and $4/11$ minutes
D) 7:58 and $2/11$ minutes
E) 8:00 exactly

23. What is the value of $\tan 30^\circ \times \tan 60^\circ$?

- A) 3 B) $\sqrt{3}/3$ C) $\sqrt{3}$ D) $1/3$ E) 1

24. For an angle θ , what is the value of $(\sin \theta + \cos \theta)^2 - 2 \sin \theta \cos \theta$?

- A) $\sin^2 \theta - \cos^2 \theta$ B) 2 C) 0 D) $2 \sin \theta \cos \theta$ E) 1

25. What is the value of $(\sin 45^\circ - \cos 45^\circ)^2 + 2 \sin 30^\circ \cos 60^\circ$?

- A) 0 B) $1/4$ C) $1/2$ D) 1 E) $3/4$

26. A straight line passes through the points (2, 5) and (6, 17). What is its slope?

- A) $1/4$ B) $1/3$ C) 3 D) 4 E) 12

27. A cyclist rides along a straight road at a constant speed. Her distance from the start is recorded as 1.8 km after 6 minutes and 3.0 km after 10 minutes. What is her speed in m/s?

- A) 3 m/s B) 5 m/s C) 18 m/s D) 50 m/s E) 300 m/s

28. Water is pumped into a tank that already contains 40 L. The volume reaches 90 L after 5 minutes and 160 L after 15 minutes. What is the average rate of inflow over the whole 15 minutes?

- A)** 7 L/min **B)** 8 L/min **C)** 8.5 L/min **D)** 10 L/min **E)** 160 L/min

Section C - Probability and combinatorics (Q29-36)

29. Two fair six-sided dice are rolled once. What is the probability that the sum of the two scores is at least 9?

- A)** $1/9$ **B)** $1/3$ **C)** $5/36$ **D)** $5/18$ **E)** $5/12$

30. Three fair six-sided dice are rolled. What is the probability that all three show the same number?

- A)** $1/216$ **B)** $1/108$ **C)** $1/72$ **D)** $1/36$ **E)** $1/6$

31. Two fair six-sided dice are rolled. What is the probability that the two scores are a 2 and a 5, in either order?

- A)** $1/36$ **B)** $1/18$ **C)** $1/9$ **D)** $1/6$ **E)** $1/3$

32. Two fair six-sided dice are rolled. What is the probability that the sum is 7 or at least one die shows a six?

- A)** $1/6$ **B)** $5/12$ **C)** $7/18$ **D)** $11/36$ **E)** $17/36$

33. A fair coin is flipped four times. What is the probability of getting exactly two heads?

- A)** $1/8$ **B)** $1/4$ **C)** $1/2$ **D)** $3/16$ **E)** $3/8$

34. An urn contains 3 red, 4 green and 5 blue balls. Two balls are drawn at random without replacement. What is the probability that at least one of them is green?

- A)** $1/3$ **B)** $3/4$ **C)** $14/33$ **D)** $16/33$ **E)** $19/33$

35. The probability that a candidate passes a maths test is $2/3$, and the probability that she passes an independent logic test is $3/4$. What is the probability that she passes exactly one of the two tests?

- A)** $1/2$ **B)** $1/6$ **C)** $1/12$ **D)** $5/12$ **E)** $11/12$

36. A fair coin is tossed and a fair six-sided die is rolled. What is the probability of obtaining tails and a score greater than 4?

- A)** $1/12$ **B)** $1/6$ **C)** $1/4$ **D)** $1/2$ **E)** $2/3$

Section D - Unit conversion and arithmetic (Q37-40)

37. A high-speed train covers 270 km in 3.0 h at a constant speed. Expressed in m/s, its speed is:

- A)** 25 m/s **B)** 75 m/s **C)** 90 m/s **D)** 324 m/s **E)** 1500 m/s

38. A swimming pool is 5.0 m long, 2.0 m wide and 1.2 m deep. What volume of water does it hold, expressed in litres?

- A)** 12 L **B)** 120 L **C)** 1200 L **D)** 12000 L **E)** 120000 L

39. A water tank is $2/3$ full. A valve then drains $1/4$ of the tank's total capacity. What fraction of the tank is still full?

- A)** $1/6$ **B)** $1/4$ **C)** $1/2$ **D)** $5/12$ **E)** $11/12$

40. A water tank is one third full. After 40 more litres are poured in, the tank is three quarters full. What is the full capacity of the tank?

- A)** 60 litres **B)** 72 litres **C)** 90 litres **D)** 96 litres **E)** 120 litres

Section E - Kinematics and dynamics (Q41-56)

- 41.** The velocity of a trolley moving along a straight track is recorded every 2 s: at 0 s it is 0 m/s, at 2 s it is 6 m/s, at 4 s it is 10 m/s, at 6 s it is 12 m/s and at 8 s it is 12 m/s. Which statement is correct?
- A)** Its acceleration between 4 s and 6 s is 1 m/s².
 - B)** Its average velocity over the whole 8 s is 10 m/s.
 - C)** It decelerates between 6 s and 8 s.
 - D)** Its acceleration is constant at 3 m/s² between 0 s and 6 s.
 - E)** It travels 96 m during the 8 s.
- 42.** In a race, a sprinter increases her speed steadily from 0 to 8.0 m/s during the first 4.0 s and then keeps running at a constant 8.0 m/s for the next 6.0 s. What is her average speed over the whole 10 s?
- A)** 1.6 m/s **B)** 4.0 m/s **C)** 4.8 m/s **D)** 6.4 m/s **E)** 8.0 m/s
- 43.** On a straight air track, a glider passes sensor P moving at 2.0 m/s and sensor Q moving at 4.0 m/s. The sensors are 6.0 m apart and the glider takes 2.0 s to travel from P to Q. Which statement is correct?
- A)** Its average velocity between the gates is 3.0 m/s and its acceleration is 2.0 m/s².
 - B)** Its average velocity between the gates is 2.0 m/s and its acceleration is 1.0 m/s².
 - C)** Its average velocity between the gates is 4.0 m/s and its acceleration is 1.0 m/s².
 - D)** Its average velocity between the gates is 6.0 m/s and its acceleration is 1.0 m/s².
 - E)** Its average velocity between the gates is 3.0 m/s and its acceleration is 1.0 m/s².
- 44.** Which of the following statements about motion graphs is correct?
- A)** For uniformly accelerated motion starting from rest, the average velocity during the acceleration is half the final velocity.
 - B)** A curved velocity-time graph means the object is moving along a circular path.
 - C)** The average velocity over any time interval always equals the mean of the initial and final velocities.
 - D)** The area under a velocity-time graph gives the acceleration.
 - E)** The gradient of a velocity-time graph gives the distance travelled.
- 45.** Which statement correctly distinguishes the tangential and the centripetal components of acceleration for an object moving in a circle?
- A)** The tangential component acts towards the centre, and the centripetal component acts along the tangent.
 - B)** Both components change only the direction of motion.
 - C)** Both components change only the speed of the object.
 - D)** The centripetal component changes the speed, and the tangential component changes the direction of motion.
 - E)** The tangential component changes the speed, and the centripetal component changes the direction of the velocity.
- 46.** A cyclist rides around a circular track of radius 20 m at a constant speed of 10 m/s. Which statement about her acceleration is correct?
- A)** It is 5.0 m/s², directed towards the centre of the track.
 - B)** It is 2.0 m/s², directed away from the centre of the track.
 - C)** It is 5.0 m/s², directed along her direction of travel.
 - D)** It is 0.50 m/s², directed along her direction of travel.
 - E)** It is zero, because her speed is constant.

- 47.** A car is driven around a circular test track while the driver steadily presses the accelerator, so the car is speeding up. At such an instant, which statement about the car's acceleration is correct?
- A)** It has a component along the motion and a component towards the centre, so the resultant points forwards and inwards.
B) It points directly towards the centre of the track.
C) It is zero, because the tangential and centripetal parts cancel.
D) It points directly away from the centre, in the direction of the outward push the driver feels.
E) It points along the direction of motion, since only a change of speed can produce acceleration.
- 48.** A car travels at 72 km/h along a straight road when the driver sees a hazard. Her reaction time is 1.0 s, after which the brakes produce a constant deceleration of 5.0 m/s² until the car stops. What total distance does the car cover between the hazard being seen and the car stopping?
- A)** 20 m **B)** 40 m **C)** 60 m **D)** 80 m **E)** 100 m
- 49.** A train travelling at 30 m/s applies its brakes and comes to rest after covering 150 m at constant deceleration. What is the magnitude of the deceleration?
- A)** 0.20 m/s² **B)** 1.5 m/s² **C)** 3.0 m/s² **D)** 5.0 m/s² **E)** 6.0 m/s²
- 50.** A crate of mass 4.0 kg is pulled along a horizontal floor by a horizontal force of 24 N. A frictional force of 8.0 N opposes the motion. What is the acceleration of the crate?
- A)** 2.0 m/s² **B)** 4.0 m/s² **C)** 6.0 m/s² **D)** 8.0 m/s² **E)** 16 m/s²
- 51.** A block of mass 2.0 kg slides down a frictionless slope inclined at 30° to the horizontal. What is its acceleration along the slope? ($g = 10 \text{ m/s}^2$, $\sin 30^\circ = 0.50$, $\cos 30^\circ = 0.87$.)
- A)** 5.0 m/s² **B)** 8.7 m/s² **C)** 10 m/s² **D)** 17.4 m/s² **E)** 20 m/s²
- 52.** A crane lifts a 50 kg load by applying a vertical upward force of 550 N. What is the upward acceleration of the load? ($g = 10 \text{ m/s}^2$.)
- A)** 0.10 m/s² **B)** 1.0 m/s² **C)** 10 m/s² **D)** 11 m/s² **E)** 21 m/s²
- 53.** An astronaut of mass 60 kg floats at rest in space and throws a tool of mass 2.0 kg so that it moves away from her at 6.0 m/s. What is the velocity of the astronaut immediately afterwards?
- A)** 0.20 m/s in the direction of the throw.
B) 0.20 m/s opposite to the direction of the throw.
C) 0.40 m/s opposite to the direction of the throw.
D) 6.0 m/s opposite to the direction of the throw.
E) 12 m/s opposite to the direction of the throw.
- 54.** A ball of mass 0.50 kg strikes a wall horizontally at 12 m/s and rebounds along its original line at 8.0 m/s. The contact with the wall lasts 0.020 s. What is the magnitude of the average force exerted by the wall on the ball?
- A)** 100 N **B)** 200 N **C)** 300 N **D)** 500 N **E)** 1000 N
- 55.** A motor lifts a 60 kg load at constant speed through a height of 10 m in 20 s. What is the useful output power of the motor? ($g = 10 \text{ m/s}^2$.)
- A)** 30 W **B)** 150 W **C)** 300 W **D)** 600 W **E)** 6000 W
- 56.** A car moves at a constant 20 m/s along a horizontal road. The total resistive force on it is 600 N. What power must the engine deliver to maintain this speed?
- A)** 30 W **B)** 600 W **C)** 6000 W **D)** 12000 W **E)** 24000 W

Section F - Fluids, thermal physics and electricity (Q57-75)

57. A diver works at a depth of 15 m in a fresh-water lake. The density of water is 1000 kg/m^3 , $g = 10 \text{ m/s}^2$, and the atmospheric pressure at the surface is 100 kPa. Which statement about the pressures on the diver is correct?

- A)** The absolute pressure at 15 m is 150 kPa, because atmospheric pressure does not act underwater.
- B)** The gauge pressure at 15 m is 250 kPa, and the absolute pressure is 350 kPa.
- C)** The gauge pressure at 15 m is 150 kPa, and the absolute pressure is 50 kPa.
- D)** The gauge pressure at 15 m is 1.5 kPa, and the absolute pressure is 101.5 kPa.
- E)** The gauge pressure at 15 m is 150 kPa, and the absolute pressure is 250 kPa.

58. An open-tube manometer contains mercury of density 13600 kg/m^3 . The mercury level in the arm open to the atmosphere stands 100 mm above the level in the arm connected to a gas supply. Atmospheric pressure is 100 kPa and $g = 10 \text{ m/s}^2$. What is the absolute pressure of the gas supply?

- A)** 13.6 kPa **B)** 86.4 kPa **C)** 100 kPa **D)** 101 kPa **E)** 113.6 kPa

59. A tank contains 0.80 m of oil of density 800 kg/m^3 floating on 1.2 m of water of density 1000 kg/m^3 . Taking $g = 10 \text{ m/s}^2$ what is the gauge pressure at the bottom of the tank?

- A)** 1.8 kPa **B)** 6.4 kPa **C)** 12 kPa **D)** 18.4 kPa **E)** 20 kPa

60. Three blocks are placed in oil of density 800 kg/m^3 . Block P floats with exactly half its volume submerged, block Q floats with one quarter of its volume submerged, and block R sinks and rests on the bottom. Which statement is correct?

- A)** Q has a density of about 200 kg/m^3 , the lowest of the three, and R is denser than the oil.
- B)** R has a density of exactly 800 kg/m^3 , the same as the oil.
- C)** P and Q must have the same density, because both are floating.
- D)** Q is denser than P, because a smaller fraction of Q is submerged.
- E)** P is the densest of the three blocks.

61. A block of weight 6.0 N and density 750 kg/m^3 floats freely in water of density 1000 kg/m^3 . What is the apparent weight of the block while it floats?

- A)** 0 N **B)** 1.5 N **C)** 2.0 N **D)** 4.5 N **E)** 6.0 N

62. How much heat must be supplied to 2.0 kg of water to raise its temperature from 20°C to 70°C ? (c of water = 4200 J/(kg K) ; ignore heat losses.)

- A)** 84 kJ **B)** 210 kJ **C)** 294 kJ **D)** 420 kJ **E)** 588 kJ

63. A 0.40 kg block of aluminium at 100°C is dropped into 0.60 kg of water at 20°C in an insulated container. (c of aluminium = 900 J/(kg K) , c of water = 4200 J/(kg K) .) What is the final equilibrium temperature?

- A)** 20°C **B)** 30°C **C)** 40°C **D)** 52°C **E)** 60°C

64. In an insulated cup, 0.50 kg of water at 12°C is mixed with hot water at 82°C , and the mixture settles at 32°C . (c of water = 4200 J/(kg K) ; ignore heat losses.) What mass of hot water was added?

- A)** 0.10 kg **B)** 0.20 kg **C)** 0.35 kg **D)** 0.50 kg **E)** 0.70 kg

65. An electric heater consumes 2.5 kWh of energy. ($1 \text{ kWh} = 3.6 \text{ MJ}$.) This energy, expressed in megajoules, is:

- A)** 0.90 MJ **B)** 2.5 MJ **C)** 9.0 MJ **D)** 25 MJ **E)** 90 MJ

66. A heating element connected across a 12 V supply carries a current of 2.0 A. How much energy does it convert to heat in 5.0 minutes?

- A)** 0.60 kJ **B)** 0.72 kJ **C)** 3.6 kJ **D)** 7.2 kJ **E)** 72 kJ

- 67.** Two identical $6.0\ \Omega$ resistors are connected, first in series and then in parallel, across a $12\ \text{V}$ battery of negligible internal resistance. Which one of the following statements is false?
- A)** The parallel arrangement dissipates 4 times as much total power as the series arrangement.
B) In the parallel arrangement the battery supplies $48\ \text{W}$.
C) In the series arrangement each resistor dissipates $6.0\ \text{W}$.
D) In the series arrangement the battery supplies $12\ \text{W}$.
E) The equivalent resistance of the parallel arrangement is $6.0\ \Omega$.
- 68.** A $6.0\ \Omega$ resistor is connected in parallel with a $3.0\ \Omega$ resistor, and this pair is connected in series with a $5.0\ \Omega$ resistor across a supply. What is the equivalent resistance of the whole network?
- A)** $2.0\ \Omega$ **B)** $5.0\ \Omega$ **C)** $7.0\ \Omega$ **D)** $11\ \Omega$ **E)** $14\ \Omega$
- 69.** A battery of negligible internal resistance supplies $24\ \text{V}$ to an $8.0\ \Omega$ resistor connected in series with a parallel combination of a $12\ \Omega$ resistor and a $6.0\ \Omega$ resistor. What current does the battery supply?
- A)** $1.5\ \text{A}$ **B)** $2.0\ \text{A}$ **C)** $2.4\ \text{A}$ **D)** $3.0\ \text{A}$ **E)** $6.0\ \text{A}$
- 70.** A $6.0\ \Omega$ resistor draws $2.0\ \text{A}$ from a battery of negligible internal resistance. When a second resistor is connected in parallel with it, the current drawn from the battery rises to $3.0\ \text{A}$. What is the resistance of the second resistor?
- A)** $3.0\ \Omega$ **B)** $6.0\ \Omega$ **C)** $12\ \Omega$ **D)** $18\ \Omega$ **E)** $36\ \Omega$
- 71.** A $4.0\ \Omega$ resistor and a $12\ \Omega$ resistor are connected in series across a $6.0\ \text{V}$ battery of negligible internal resistance. A wire of negligible resistance is then connected across the $12\ \Omega$ resistor. What current does the battery supply once the wire is connected?
- A)** $0.375\ \text{A}$ **B)** $0.50\ \text{A}$ **C)** $1.5\ \text{A}$ **D)** $3.0\ \text{A}$ **E)** $6.0\ \text{A}$
- 72.** Intensity measures how fast a wave delivers energy to a surface, per unit area of that surface. The SI unit in which intensity is expressed is:
- A)** W/m^3 **B)** J/m^2 **C)** W/m **D)** J/s **E)** W/m^2
- 73.** A small lamp is treated as a point source radiating uniformly in all directions. At $2.0\ \text{m}$ from the lamp the intensity of its radiation is $36\ \text{W/m}^2$. What is the intensity at $6.0\ \text{m}$ from the lamp?
- A)** $4.0\ \text{W/m}^2$ **B)** $6.0\ \text{W/m}^2$ **C)** $12\ \text{W/m}^2$ **D)** $18\ \text{W/m}^2$ **E)** $36\ \text{W/m}^2$
- 74.** In a heated room, a radiator warms the air beside it. This air rises, cooler air replaces it, and the resulting circulating flow carries thermal energy around the room. This circulating flow transports the energy mainly by:
- A)** conduction through the metal of the radiator, the best heat conductor present
B) evaporation, because the flow carries water vapour that releases heat where it condenses
C) radiation, because the hot radiator emits infrared rays that travel in straight lines
D) conduction, because neighbouring air molecules pass energy on by direct collisions
E) convection, because the warmed air expands, becomes less dense and rises within the flow
- 75.** A vacuum flask keeps a drink hot. It has double walls with silvered surfaces and a vacuum between them, closed at the top by a stopper. Which statement about how the flask limits heat loss is correct?
- A)** The silvered walls mainly reduce conduction through the walls.
B) The vacuum blocks convection only, because conduction can still occur through a vacuum.
C) The stopper works mainly by reflecting radiation back into the drink.
D) The vacuum mainly blocks radiation, and the silvering mainly blocks convection.
E) The vacuum blocks both conduction and convection, and the silvered surfaces cut radiation losses.

Section G - Logic and argument analysis (Q76-95)

76. In a library, every manuscript in the special archive is catalogued. Some of the archive's manuscripts are illustrated, and no illustrated item in the archive may leave the building. Which of the following must be true?

- A)** Some of the archive's illustrated manuscripts are catalogued.
- B)** Every catalogued item in the library is illustrated.
- C)** Every item that may leave the building is catalogued.
- D)** Some catalogued items in the library may leave the building.
- E)** No catalogued item in the archive is illustrated.

77. At a tennis club, only members with a current season pass may book the courts, and all club rules are followed. Nadia booked a court this week. Some members with a season pass also take coaching lessons. Which of the following must be true?

- A)** Nadia holds a current season pass.
- B)** Every member who books the courts takes coaching lessons.
- C)** Some members who book the courts take coaching lessons.
- D)** Every member with a season pass books the courts.
- E)** Nadia takes coaching lessons.

78. Every swimmer who won a medal this season trains at the Riverside pool, and nobody who trains at the Riverside pool pays a joining fee. Which of the following must be true?

- A)** Some people who train at the Riverside pool did not win a medal this season.
- B)** The Riverside pool charges no joining fee to anyone.
- C)** Some swimmers who pay a joining fee won a medal this season.
- D)** Every swimmer who trains at the Riverside pool won a medal this season.
- E)** No swimmer who won a medal this season pays a joining fee.

79. Some players in the squad are left-footed. No left-footed player takes the corner kicks, and every player in the squad is a professional. Which of the following must be true?

- A)** Some professionals do not take the corner kicks.
- B)** Every player who takes the corner kicks is right-footed.
- C)** No professional takes the corner kicks.
- D)** Every player who takes the corner kicks is in the squad.
- E)** Some left-footed players are not professionals.

80. Every baker at the Saturday market arrives before dawn, and some bakers at the market sell rye bread. Which of the following must be true?

- A)** Everyone who arrives before dawn sells rye bread.
- B)** Some people who arrive before dawn sell rye bread.
- C)** All sellers of rye bread at the market are bakers.
- D)** No baker at the market sells rye bread.
- E)** Some bakers at the market arrive after dawn.

81. Every member of the rowing club can swim, and no member of the rowing club is under sixteen. Tara is a member of the rowing club. Which of the following must be true?

- A)** Some people under sixteen can swim.
- B)** Everyone who can swim is a member of the rowing club.
- C)** Tara is under sixteen.
- D)** Only swimmers may join the rowing club.
- E)** Tara can swim and is not under sixteen.

- 82.** About a postponed football match, Ria says: 'The match was postponed because of rain.' Tom says: 'The match was postponed because of rain, and Ria is lying.' Exactly one of the two always tells the truth and the other always lies. Which of the following must be true?
- A)** Tom tells the truth and rain caused the postponement.
 - B)** Ria tells the truth and rain caused the postponement.
 - C)** Ria tells the truth and the postponement had another cause.
 - D)** Tom tells the truth and the postponement had another cause.
 - E)** Both of them always tell the truth.
- 83.** Three pupils, Amara, Ben and Cara, are asked about a missing lab key. Each of them either always tells the truth or always lies. Amara says: 'Ben is lying.' Ben says: 'Cara is lying.' Cara says: 'Both Amara and Ben are lying.' Which of the following is correct?
- A)** Amara and Cara lie and Ben tells the truth.
 - B)** The three statements allow more than one consistent assignment.
 - C)** Ben and Cara lie and Amara tells the truth.
 - D)** Amara and Ben tell the truth and Cara lies.
 - E)** All three tell the truth.
- 84.** Three teachers are questioned about a borrowed stopwatch. Exactly one of them is lying and the other two are telling the truth. Priya says: 'I did not take the stopwatch.' Quinn says: 'Priya is lying.' Ravi says: 'Quinn is lying.' Which of the following must be true?
- A)** Priya took the stopwatch.
 - B)** All three are telling the truth.
 - C)** Quinn took the stopwatch.
 - D)** Ravi took the stopwatch.
 - E)** Priya did not take the stopwatch.
- 85.** After a match, the scoreboard was found broken. Exactly one of the three candidates for team captain always tells the truth and the other two always lie. Dana says: 'Eli broke the scoreboard.' Eli says: 'I did not break the scoreboard.' Fran says: 'Dana is lying.' Which of the following must be true?
- A)** Eli tells the truth and Fran broke the scoreboard.
 - B)** Fran tells the truth and Dana broke the scoreboard.
 - C)** Eli tells the truth and Dana broke the scoreboard.
 - D)** Dana tells the truth and Eli broke the scoreboard.
 - E)** Fran tells the truth and Eli broke the scoreboard.
- 86.** At a lost property office, exactly one of Pia's two statements is true and exactly one of Sam's two statements is true. The clerk confirms the blue bag was collected on exactly one day. Pia says: 'The bag was collected on Monday.' and 'The bag was not collected on Wednesday.' Sam says: 'The bag was collected on Wednesday.' and 'The bag was not collected on Monday.' Which of the following must be true?
- A)** The bag was collected on Wednesday.
 - B)** The bag was collected on Monday.
 - C)** Sam made two true statements.
 - D)** Pia made two false statements.
 - E)** The bag was collected on a day that is neither Monday nor Wednesday.
- 87.** A notice at the swimming pool reads: 'If the pool is at capacity, new swimmers are turned away.' Which of the following is logically equivalent to the notice?
- A)** If the pool is not at capacity, new swimmers are turned away.
 - B)** If new swimmers are not turned away, the pool is not at capacity.
 - C)** If new swimmers are turned away, the pool is at capacity.
 - D)** If the pool is not at capacity, new swimmers are not turned away.
 - E)** New swimmers are turned away only if the pool is at capacity.

- 88.** The coach tells the squad: 'If a player attends every training session this month, she will be selected for the final.' Which of the following is logically equivalent to the coach's statement?
- A)** A player who attends at least one training session this month will be selected for the final.
 - B)** A player will be selected for the final only if she attends every training session this month.
 - C)** If a player does not attend every training session this month, she will not be selected for the final.
 - D)** If a player is not selected for the final, she did not attend every training session this month.
 - E)** If a player is selected for the final, she attended every training session this month.
- 89.** Under the library's loan rule, a book may be borrowed for a full term only if it is on the reserve list. Which of the following must also be true if the rule is obeyed?
- A)** A book not on the reserve list may not be borrowed for a full term.
 - B)** Every book on the reserve list is borrowed for a full term.
 - C)** A book not on the reserve list may be borrowed for a full term.
 - D)** A book on the reserve list may be borrowed for a full term.
 - E)** Only books on the reserve list may be borrowed at all.
- 90.** A ferry company's policy states that if a sailing is delayed, meal vouchers are issued, and that if meal vouchers are issued, the terminal canteen stays open late. Which of the following is logically equivalent to the policy taken as a whole?
- A)** Every delayed sailing leaves passengers hungry.
 - B)** If a sailing is delayed, it is because the canteen plans to stay open late.
 - C)** If the canteen does not stay open late, the sailing was not delayed.
 - D)** If a sailing is not delayed, the canteen does not stay open late.
 - E)** If the canteen stays open late, the sailing was delayed.
- 91.** A security manual states: 'The side gate stays locked if and only if the alarm is armed.' Which of the following is logically equivalent to this statement?
- A)** The alarm is armed whenever the side gate is unlocked.
 - B)** If the alarm is disarmed, the side gate is locked.
 - C)** If the alarm is armed, the side gate is unlocked.
 - D)** If the side gate is locked, the alarm is disarmed.
 - E)** The side gate is unlocked exactly when the alarm is disarmed.
- 92.** A council reports that streets repaved with a new porous surface drain rainwater quickly and form far fewer puddles, and concludes that repaving will reduce road accidents in which vehicles slip. The conclusion depends on assuming which of the following?
- A)** Accidents in which vehicles slip are usually caused by surface water, and the porous surface does not make roads slippery in other ways.
 - B)** Drivers on repaved streets tend to be more cautious than other drivers.
 - C)** The porous surface costs less per kilometre than standard asphalt.
 - D)** The council intends to repave most streets within five years.
 - E)** Standard asphalt forms more puddles than the porous surface.
- 93.** At a gym, members on the flexible plan attend twice as often as members on the standard plan and report larger fitness gains. The manager concludes that the flexible plan itself causes the better fitness. Which of the following, if true, most weakens the manager's conclusion?
- A)** The flexible plan costs more per month than the standard plan.
 - B)** Members choose their own plan, and most members who were already training hard before joining picked the flexible plan.
 - C)** Standard plan members report busier weekday schedules.
 - D)** Attendance at the gym is recorded automatically at the entrance.
 - E)** Both groups took the same fitness assessment at the start and at the end of the study.

94. A ferry operator installed automatic ticket scanners on two of its six routes. Boarding time on those two routes fell by a third within two months and stayed flat on the other four. The operator concludes the scanners caused the faster boarding. Which of the following, if true, most strengthens this conclusion?

- A)** Passengers on the scanner routes said in a survey that they liked the scanners.
- B)** Both scanner routes use ferries of the same model.
- C)** Over the same two months the operator ran an advertising campaign on all six routes.
- D)** When the scanners were out of service for two days, boarding times on those routes briefly returned to their old level and fell again after repairs.
- E)** Staff numbers were also increased at the terminals of the two scanner routes.

95. Last year a sports centre cut its evening swim sessions from five per week to three. Over the year, total attendance at evening swims stayed almost unchanged, and the waiting list for evening swims grew longer. Which of the following is most strongly supported by this information?

- A)** Some swimmers gave up swimming in the evenings.
- B)** The centre should restore the two cancelled sessions.
- C)** On average, each remaining evening session now accommodates more swimmers than before.
- D)** The waiting list caused the sessions to be cut.
- E)** Attendance at morning swims must have increased.

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	C	2	B	3	E	4	C	5	D
6	A	7	D	8	A	9	B	10	B
11	C	12	D	13	C	14	B	15	C
16	E	17	C	18	A	19	A	20	D
21	C	22	C	23	E	24	E	25	C
26	C	27	B	28	B	29	D	30	D
31	B	32	B	33	E	34	E	35	D
36	B	37	A	38	D	39	D	40	D
41	A	42	D	43	E	44	A	45	E
46	A	47	A	48	C	49	C	50	B
51	A	52	B	53	B	54	D	55	C
56	D	57	E	58	E	59	D	60	A
61	A	62	D	63	B	64	B	65	C
66	D	67	E	68	C	69	B	70	C
71	C	72	E	73	A	74	E	75	E
76	A	77	A	78	E	79	A	80	B
81	E	82	B	83	A	84	E	85	D
86	E	87	B	88	D	89	A	90	C
91	E	92	A	93	B	94	D	95	C